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CSC 360 Assignment #3

- 4.2. Discover ambiguities or omissions in the following statement of the requirements for part of a ticket-issuing system:

An automated ticket machine sells rail tickets. Users select their destination and input a credit card and a personal identification number. The rail ticket is issued and their credit card account charged. When the user presses the start button, a menu display of potential destinations is activated, along with a message to the user to select a destination and the type of ticket required. Once a destination has been selected, the ticket price is displayed and customers are asked to input their credit card. Its validity is checked, and the user is then asked to input his or her personal identifier (PIN). When the credit transaction has been validated, the ticket is issued.

Reading the following statement has gotten me to identify a few ambiguities. First of all, there is no response from the program when using it. How is the user supposed to know whether or not their card was charged? How are they supposed to know whether or not their credit card is valid? Confirmation messages like these will help the user not be so confused when going through the ticket machine. Moreover, there is no mention of error handling within the machine. If the user accidentally chooses the wrong destination they want to go to, there is no way for them to go back to choose the right one. Furthermore, there are more details needed regarding selecting a destination. The details seem very vague and there is no way of knowing if there are multiple types of tickets or not. These are a few of the ambiguities that I found in the following statement.

- 4.4. Write a set of non-functional requirements for the ticket-issuing system, setting out its expected reliability and response time.
- The machine goes back to the main screen if nothing happens within a 45 second increment.
- It should load up each screen within 5 seconds when a button is pressed.
- When selecting a destination, it gives you a ratio of how many seats are available. Also, it will show if that destination is open to be traveled to that day. If not, there is the ability to buy a ticket for later in the week.
- All transactions are logged securely within the system.
- In the process of a crash, the entire process in the machine should be terminated so there is no charge to the card without a ticket.
- 4.5. Using the technique suggested here, where natural language descriptions are presented in a standard format, write plausible user requirements for the following functions: An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader, then specifies the amount of fuel required. The fuel is delivered and the customer’s account debited.
- The system shall authenticate users through a credit card reader by swiping their cards within the machine.
- The system shall allow the ability for users who are identified to specify what kind, and how much fuel they want.

- Once selected, the user shall confirm their transaction and the charge should be applied to their card at this point.
- Upon confirmation, the fuel shall be dispensed.
- Once all bought fuel is dispensed, the system shall print a confirmation message for the user to confirm that the entire process worked as expected.
- Before the user's card is charged, if the system is left alone for more than 30 seconds, the entire system shall terminate and go back to the starting screen.
- If there is a crash after the charge is applied, the system shall not charge the user so they can try again.